SCHOOL OF INFORMATION AND PHYSICAL SCIENCES
BUSINESS INTELLIGENCE FOR THE ENTERPRISE
B&B BUSINESS CASE

Assessment 2 : **B&B Business Case - Group Project - Part B**

Course : **INFO6090**
Campus : **Callaghan**
Lab Timings : **Wednesday 5.00 PM – 7.00 PM**

Group B-Members:

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BEAN & BLOOM STRATEGIC INSIGHTS REPORT

EXECUTIVE SUMMARY:

Bean and Bloom's 2022 national performance across 10 stores generated $4,335,793 in revenue from 20,000 transactions, achieving a robust average transaction value of $216.79 and an exceptional loyalty participation rate of 97.4%.

However, the current 12.5% discount model triggered by purchases of 5 plus distinct items cost $414,871 that is 9.57% of revenue while showing no meaningful impact on incremental sales or visit frequency. Loyalty members average 30.6 visits and $7,018 in spend, compared to 2.9 visits and $224 for non-members, but this gap is driven by baseline engagement, not discount incentives.

Evidence suggests customers are gaming the system by bundling low-value items, inflating transaction counts without boosting margins. Store performance reveals stark concentration: Wagga Wagga leads with $707,116 which is 16.3% of total, while the top 5 stores account for 69.2% of revenue. In contrast, Wollongong $332,040, Dubbo $351,151, and West Wyalong $368,611 lag significantly, indicating operational inefficiencies or market misalignment.

Product analysis highlights 1kg House Blend Beans as the undisputed leader, dominating 7 of the top 10 product-store combinations, while multiple variations of Flat White, Chia Pudding, Long Black, Banana Bread, and Croissant underperform across the board, creating menu bloat and inventory complexity without revenue benefit.

Forecasting models project a 19% revenue decline in 2023 to $3,511,562 monthly average is $320,433 vs. $395,888 in 2022, driven by market saturation, loyalty fatigue, competitive pressure, and absence of growth initiatives.

Model reliability averages $R^2 = 0.304$, with Dubbo showing strong predictability R^2 0.507 and Newcastle high volatility R^2 is 0.027. To reverse this trajectory, we recommend immediate transition to a points-based loyalty system 1 point per $1 spent, redeemable on curated feature products, expected to recover 3-4% margin $130,000–$173,000 annually, eliminate gaming, and foster genuine engagement.

Concurrently, streamline the menu to 8–10 core items by removing the bottom 3–5 items, focus inventory on proven winners like House Blend Beans, and introduce 2–3 premium replacements. Adopt a tiered store strategy: protect and expand top performers Wagga Wagga, Broken Hill, Maitland, optimize middle-tier locations, and strategically review or consolidate underperforming sites.

Key Findings at a Glance:

Metric	Value	Insight
Total Annual Revenue	$4335793.08	Baseline performance established
Total Transactions	20,000	Consistent customer traffic
Average Transaction Value	$216.79	Strong basket size
Total Discounts Given	$414871.03	9.6% of revenue
Average Projected 2023 Revenue	$320433	-19% decline

Critical Recommendation

We strongly recommend transitioning from the current 12.5% discount model to a points-based rewards system**.** Our analysis shows the current discount structure is not driving incremental sales growth sufficient to justify its cost, and market trends suggest a decline in 2023 revenue if operational changes are not implemented.

1. STORE PERFORMANCE ANALYSIS

1.1 Top Performing Stores:

Wagga Wagga with Store 9 leads in all key metrics Revenue: $707,116 which is 16.3% of total Transactions with $3,221 which is 16.1% of total and Average transaction value is $219.53

Top 5 Stores by Revenue:

1. Wagga Wagga: $707,116

2. Broken Hill: $474,238

3. Maitland: $472,048

4. Newcastle: $432,668

5. Lismore: $423,738

These 5 stores generate 69.2% of total revenue, indicating concentration risk.

1.2 Underperforming Stores:

Bottom 3 Stores:

1. Wollongong: $332,040 which is 7.7% of total

2. Dubbo: $351,151 which is 8.1% of total

3. West Wyalong: $368,611 which is 8.5% of total

Key Concern: These stores show lower customer traffic and transaction values, suggesting market fit or operational challenges.

1.3 Store Performance Recommendations:

Store Category	Action Required
Top 3 (Wagga Wagga, Broken Hill, Maitland)	Protect & Grow: Invest in expansion capacity, staff training programs
Middle 4 (Newcastle, Lismore, Port Macquarie, Tammworth)	Optimize: Implement best practices from top performers
Bottom 3 (West Wyalong, Dubbo, Wollongong)	Strategic Review: Consider restructuring, marketing push, or consolidation

2. PRODUCT PERFORMANCE INSIGHTS

2.1 Nationwide Best Sellers (Top 3 Products)

1. **1kg House Blend Beans** - Dominates across ALL stores
   -Represents 7 of top 10 product-store combinations
   -Consistent performer regardless of location
   Category: Merchandise - premium coffee beans
2. **Espresso**- High volume, low margin
3. **Almond Milk Upgrade**- Consistent daily traffic driver

2.2 Nationwide Underperformers (Bottom Quartile)

Products Recommended for Removal:

Product	Total Revenue	Status	Recommendation
Flat White	$ 57446.64	8 variations underperforming	Consolidate to 2-3 variations max
Chia Pudding	$72869.01	7 variations underperforming	Remove bottom 5 variations
Long Black	$85748.39	6 variations underperforming	Consolidate significantly
Banana Bread	$139848.09	Low appeal	Consider replacement
Croissant	$131622.79	Low volume	Consider replacement

Key Insight: Product menu is overly complex with too many variations of the same core items This creates operational complexity without revenue benefit.

2.3 Product Recommendations:

1. **Immediate Actions:**
    - Remove bottom 3 - 5 underperforming products.
    - Reduce product variations consolidate 15 products down to 8-10 core offerings.
    - Focus inventory on 1kg House Blend Beans which is clear winner

2. **Strategic Actions:**
    - Introduce 2-3 new premium products to replace removed items.
    - Create seasonal rotation for variety without complexity.
    - Promote underperformers through "feature product" program in new points system.

3. LOYALTY PROGRAM EFFECTIVENESS ANALYSIS

3.1 Current State Assessment:

The current 12.5% discount model shows concerning inefficiencies:

Customer Type	Avg Visits	Avg Spend	Total Discount Received
Loyalty Members (Yes)	30	$7,018	$414,871
Non-Members (No)	2	$224	$ 0

3.2 Critical Problems Identified:

Problem 1: Discount Erosion - Total discounts: $414,871 which is 9.57% of revenue This represents a significant margin sacrifice, No evidence that discounts drive incremental frequency beyond baseline loyalty

Problem 2: Low Incremental Value - Loyalty members spend $7,018 versus non-members $224. But this is primarily due to visit frequency 30.6 versus 2.9, not discount impact. The 12.5% discount is being given to customers who would likely visit anyway

Problem 3: Transaction Structure Gaming - Customers need 5+ distinct items to qualify, this may encourage lower-value item bundling rather than premium purchases. Evidence is Average items per transaction 30.2 suggests bulk buying low-cost items.

3.3 Loyalty Program Recommendations:

RECOMMENDED: Transition to Points-Based System

Feature	Current Model	Recommended Model
Incentive Type	12.5% discount on 5+ items	1 point per $1 spent
Redemption	Immediate discount	Monthly curated feature products
Customer Behavior	Bundle low-value items	Focus on spend, not item count
Business Impact	9.57% margin erosion	Controlled redemption costs
Engagement	Transaction-level	Ongoing relationship building

Expected Benefits:
1. **Cost Control:** Redemption rate typically 60-70% of points earned versus 100% current discount.
2. **Margin Improvement:** Recover 3-4% margin, estimated $130,000-$173,000 annually.
3. **Customer Behavior:** Shift from gaming to genuine engagement.
4. **Strategic Flexibility:** Feature slow-moving products as redemption options.
5. **Data Insights:** Points balances create ongoing engagement opportunities.

4. FORECASTING & PREDICTIVE INSIGHTS

4.1 12-Month Revenue Forecast:

Using time series analysis with exponential smoothing and seasonal adjustments:

Nationwide Projection:
Forecasted monthly average : $ 320432.98
Avg_Historical_Revenue (12 months) : $395888
Expected decline : 19.0% vs. 2022

4.2 Store-Level Forecast Insights:

Model Performance (LinearRegression+Trend+Seasonality): -

Average R^2 score : 0.304 moderate predictive power
Best forecast accuracy : Dubbo R^2 = 0.507
Lowest accuracy : Newcastle with R^2 = 0.027 - high volatility

StoreLocation	Month	ForecastRevenue
Newcastle	1/01/2023	45360.34
Newcastle	1/02/2023	37592.74
Newcastle	1/03/2023	48454.96
Newcastle	1/04/2023	42759.34
Newcastle	1/05/2023	41764.15
Newcastle	1/06/2023	37417.68
Newcastle	1/07/2023	36568.47
Newcastle	1/08/2023	43988.11
Newcastle	1/09/2023	39794.16
Newcastle	1/10/2023	36999.36
Newcastle	1/11/2023	45321.27
Newcastle	1/12/2023	40031.06
Maitland	1/01/2023	41870.02
Maitland	1/02/2023	38358.34
Maitland	1/03/2023	32084.44
Maitland	1/04/2023	38717.48
Maitland	1/05/2023	35655.87
Maitland	1/06/2023	32046.17
Maitland	1/07/2023	38091.78
Maitland	1/08/2023	30991.89
Maitland	1/09/2023	32767.11
Maitland	1/10/2023	30213.69
Maitland	1/11/2023	34123.71
Maitland	1/12/2023	32185.84
Tammworth	1/01/2023	34020.19
Tammworth	1/02/2023	23382.98
Tammworth	1/03/2023	24107.82
Tammworth	1/04/2023	26428.25
Tammworth	1/05/2023	27109.15
Tammworth	1/06/2023	20740.18
Tammworth	1/07/2023	26228.3
Tammworth	1/08/2023	22934.83
Tammworth	1/09/2023	21710.96
Tammworth	1/10/2023	25904.2
Tammworth	1/11/2023	24301.18
Tammworth	1/12/2023	22557.56
Lismore	1/01/2023	32167.05
Lismore	1/02/2023	23830.49
Lismore	1/03/2023	25288.1
Lismore	1/04/2023	29890.71

Lismore	1/05/2023	21948.28
Lismore	1/06/2023	18915.09
Lismore	1/07/2023	28953.39
Lismore	1/08/2023	24859.12
Lismore	1/09/2023	22394.39
Lismore	1/10/2023	24181.25
Lismore	1/11/2023	21343.13
Lismore	1/12/2023	19570.89
Port Macquarie	1/01/2023	28938.98
Port Macquarie	1/02/2023	28076.62
Port Macquarie	1/03/2023	24205.61
Port Macquarie	1/04/2023	33016.72
Port Macquarie	1/05/2023	28457.54
Port Macquarie	1/06/2023	26728.44
Port Macquarie	1/07/2023	34501.59
Port Macquarie	1/08/2023	29597.74
Port Macquarie	1/09/2023	25196.25
Port Macquarie	1/10/2023	32369.35
Port Macquarie	1/11/2023	35625.4
Port Macquarie	1/12/2023	34931.03
West Wyalong	1/01/2023	29227.56
West Wyalong	1/02/2023	24326.55
West Wyalong	1/03/2023	27041.35
West Wyalong	1/04/2023	38584.23
West Wyalong	1/05/2023	29548.67
West Wyalong	1/06/2023	32001.38
West Wyalong	1/07/2023	27607.82
West Wyalong	1/08/2023	26206.03
West Wyalong	1/09/2023	29899.58
West Wyalong	1/10/2023	30023.44
West Wyalong	1/11/2023	31943.34
West Wyalong	1/12/2023	24405.47
Dubbo	1/01/2023	27194.26

Dubbo	1/02/2023	28414.88
Dubbo	1/03/2023	24597.32
Dubbo	1/04/2023	26429.2
Dubbo	1/05/2023	30037.56
Dubbo	1/06/2023	30571.63
Dubbo	1/07/2023	36321.97
Dubbo	1/08/2023	32817.41
Dubbo	1/09/2023	30465.75
Dubbo	1/10/2023	36578.45
Dubbo	1/11/2023	36532.87
Dubbo	1/12/2023	37730.81
Wollongong	1/01/2023	28488.15
Wollongong	1/02/2023	19111.83
Wollongong	1/03/2023	26179.84
Wollongong	1/04/2023	21296.86
Wollongong	1/05/2023	22568.29
Wollongong	1/06/2023	19796.24
Wollongong	1/07/2023	29458.79
Wollongong	1/08/2023	20142.4
Wollongong	1/09/2023	15064.59
Wollongong	1/10/2023	19163.92
Wollongong	1/11/2023	25259.23
Wollongong	1/12/2023	18123.98
Wagga Wagga	1/01/2023	54907.84
Wagga Wagga	1/02/2023	39065.1
Wagga Wagga	1/03/2023	32136.35
Wagga Wagga	1/04/2023	42696.4
Wagga Wagga	1/05/2023	39523.35
Wagga Wagga	1/06/2023	34541.1
Wagga Wagga	1/07/2023	46319.4
Wagga Wagga	1/08/2023	38970.84
Wagga Wagga	1/09/2023	32753.73
Wagga Wagga	1/10/2023	37087.18
Wagga Wagga	1/11/2023	35897.96
Wagga Wagga	1/12/2023	29445.64
Broken Hill	1/01/2023	44682.21
Broken Hill	1/02/2023	44231.8
Broken Hill	1/03/2023	46320.84
Broken Hill	1/04/2023	43756.04
Broken Hill	1/05/2023	49228.4
Broken Hill	1/06/2023	46542.53
Broken Hill	1/07/2023	47573.99

Broken Hill	1/08/2023	40467.59
Broken Hill	1/09/2023	38062.69
Broken Hill	1/10/2023	39092.05
Broken Hill	1/11/2023	40734.82
Broken Hill	1/12/2023	45699.28
NATIONWIDE	1/01/2023	370104.3
NATIONWIDE	1/02/2023	306380
NATIONWIDE	1/03/2023	308273.5
NATIONWIDE	1/04/2023	343698.9
NATIONWIDE	1/05/2023	324410.2
NATIONWIDE	1/06/2023	297583.9
NATIONWIDE	1/07/2023	353797
NATIONWIDE	1/08/2023	311398.6
NATIONWIDE	1/09/2023	289023.2
NATIONWIDE	1/10/2023	310478.9
NATIONWIDE	1/11/2023	329198
NATIONWIDE	1/12/2023	300849.2

Explanation:

Step	Description
1. Data Loading & Cleaning	Reads the 28-column CSV (no header), assigns correct column names, converts ConvertedSaleDate to datetime and RowTotal_Num to numeric. Drops invalid rows.
2. Monthly Aggregation	Groups by StoreID, StoreLocation, and **start of month** (MS) to compute total revenue per store per month.
3. Feature Engineering	Adds: • MonthNum (1 to 12+) for trend • Quarter (1–4) • MonthOfYear (1–12) for seasonality
4. Model Training (Per Store)	Fits a **Linear Regression** on [MonthNum, Quarter, MonthOfYear] → predicts base revenue.
5. Trend Adjustment	Calculates **recent 3-month vs prior 3-month change** and adds **30% of that momentum** to future predictions.
6. Seasonality Scaling	Applies **historical monthly average / overall average** as a **seasonal multiplier** to each forecast month.
7. Forecast Generation	Produces 12 future months starting from the **last observed month + 1**. Ensures non-negative values.
8. Nationwide Roll-up	Forecasts national total **directly** (not sum of stores) for consistency.
9. Evaluation & Output	Computes **MAE, RMSE, R^2** per store. Saves CSVs and generates **clean grid plot + model performance dashboard**.

4.3 Forecast Implications:

WARNING: The 19% projected decline is driven by:

Market saturation - Current customer base may be near capacity
No growth initiatives - Projection assumes status quo operations
Competitive pressure - Market trends suggest increasing competition
Loyalty fatigue - Current discount model not driving retention

MITIGATION STRATEGIES:
1. Implement points-based loyalty system.
2. Launch targeted marketing campaigns in underperforming stores.
3. Introduce 2-3 premium products to drive average transaction value.
4. Focus on customer retention programs reduce churn from current 48% one-time visitors.

5. ASSUMPTIONS & LIMITATIONS

5.1 Key Assumptions:

Data Quality: The data quality assumes cleaned data accurately represents actual business operations.
Historical Patterns: The historical patterns forecasts assume 2022 patterns will continue without intervention.
Market Conditions: The market conditions does not account for major economic shifts or new competitors.
Loyalty Behavior: The loyalty behaviour assumes current customer segments will respond similarly to new points system.
Seasonal Adjustments: The seasonal adjustments are applied based on 2022 monthly variations.

5.2 Limitations:

Forecast Accuracy: The forecast accuracy with R^2 scores of 0.304 average indicate moderate uncertainty.
External Factors: The external factors cannot predict economic recession, supply chain disruptions, or competitor actions.
Points System Adoption: The point system adoption recommendation assumes 70-80% customer participation in new system.
Employee Retention: The employee retention assumes current employee base remains stable.
Product Testing: The product testing is that new product recommendations require market testing before full rollout.

6. CONCLUSION

Bean and Bloom stands at a critical juncture. While the business has strong fundamentals loyal customer base, proven products 1kg House Blend Beans and profitable top-tier stores the current trajectory projects a 19% revenue decline in 2023.

Our three strategic imperatives are transform the loyalty model from a costly discount program to a strategic points-based system that controls costs while maintaining engagement. Simplify and strengthen the product portfolio by eliminating underperformers and focusing on proven winners. Make decisive investment choices by protecting top performers, optimizing middle-tier stores, and strategically reviewing and potentially consolidating bottom-tier locations.

The cost of inaction is clear is without intervention, Bean and Bloom will lose nearly $716321 Monthly in revenue in 2023. With the recommended changes, we project the business can stabilize and position for 5-10% growth instead.

The Board should approve the points-based loyalty system implementation, product portfolio optimization, tiered investment strategy and tier 3 store strategic review with consolidation authority.

INFO6090 Weekly Team Meeting Progress and Planning Sheet

Team: **Group-B**

Semester Weeks 08-11 (19.10.2025 - 15.11.2025)

Member ID	Members(given, Family Name)	Members Present (ID)
1	Hari Krishna Jadda	c3511138
2	Madhavi Latha Tripuram	c3528532
3	Dhimanshi Piripun Jayasena	c3503338
4	Sankeerth Sappa	c3508025

Task ID (keep old ID's)	Task Description	Members Assigned (IDs)	Estimated Completion Date	Work In Progress (WIP) Due Date(s)	Work in Progress Uploaded Status	Task Status	Comments
T01	Project Planning & National Dataset Analysis	1,2,3,4	22.10.2025	22.10.2025	Yes	Completed	Initial team meeting to analyze Part B requirements, review national dataset structure, and collaboratively plan task distribution across all members.
T02	National Data Model Design (Star Schema for 10 Stores)	1,2,3,4	28.10.2025	27.10.2025	Yes	Completed	Collaborative design of scalable star schema. All members contributed to schema design, business rule validation, and ERD creation.
T03	ETL Process Development - Data Extraction & Initial Staging	1,4	30.10.2025	30.10.2025	Yes	Completed	Data extraction from national dataset and initial staging table creation. Handled larger dataset volume and preliminary data profiling.
T04	ETL Process Development - Data Cleaning & Transformation	1,2,3,4	02.11.2025	01.11.2025	Yes	Completed	Comprehensive data cleaning across all 10 stores. Team divided store-specific cleaning tasks, collaborated on standardization rules, and implemented SCD Type 2 for price history.
T05	ETL Documentation & Data Quality Report	3,4	03.11.2025	03.11.2025	Yes	Completed	Comprehensive documentation of ETL process, transformation logic, data quality issues, and validation results. All members provided input on assumptions and business rules.
T06	SQL Development - Store Performance Comparison Queries	2,3	05.11.2025	04.11.2025	Yes	Completed	Developed queries for revenue, customer traffic, and transaction analysis across 10 stores. Other team members contributed to query validation and optimization.
T07	SQL Development - Product Performance & Loyalty Analysis	2,3	06.11.2025	06.11.2025	Yes	Completed	Queries for top/bottom products per store, loyalty segmentation, and discount effectiveness. Team collaborated on analytical requirements and result interpretation.
T08	SQL Development - Trend Analysis & Forecasting Support Queries	1,2	07.11.2025	07.11.2025	Yes	Completed	Time series queries extracting historical sales data for forecasting models. All members validated query outputs for completeness and accuracy.
T09	Forecasting Model Development & Implementation	1,2,3,4	09.11.2025	08.11.2025	Yes	Completed	Team collaboration on forecasting methodology selection. All members contributed to model validation, assumption testing, and business interpretation.
T10	Forecasting Documentation & Visualization	1,3	10.11.2025	10.11.2025	Yes	Completed	Documentation of forecasting methodology and creation of forecast visualizations. Team review ensured clarity for business stakeholders.
T11	Dashboard Development - Data Preparation & Design	2,4	11.11.2025	10.11.2025	Yes	Completed	Power BI dashboard setup, data source connection, and initial design. All members provided feedback on layout and visualization requirements.
T12	Dashboard Development - Visualization & Interactivity Implementation	1,2	12.11.2025	11.11.2025	Yes	Completed	Built interactive features, filters, and visualizations. Team tested dashboard functionality and provided user experience feedback for refinement.
T13	Strategic Insights Report - Content Development	3,4	13.11.2025	12.11.2025	Yes	Completed	Draft executive summary and strategic recommendations. All members contributed insights from their analytical domains and reviewed drafts for accuracy.
T14	Strategic Insights Report - Final Writing & Integration	1,2,3,4	13.11.2025	14.11.2025	Yes	Completed	Finalized 3-page report synthesizing all findings. Collaborative review ensured consistent messaging across all analytical components and deliverables.
T15	Quality Assurance & Cross-Deliverable Consistency Check	1,2,3,4	14.11.2025	15.11.2025	Yes	Completed	Comprehensive QA with all members validating consistency between report findings, dashboard visualizations, SQL outputs, and forecasts. Verified all calculations.
T16	Peer Review Discussion & Completion	1,2,3,4	16.11.2025	16.11.2025	Yes	Completed	Collaborative discussion of contributions. Team agreed on equal 25% distribution recognizing balanced workload and genuine collaboration across all deliverables.
T17	Final Submission Package Preparation & Upload	1,2,3,4	16.11.2025	16.11.2025	Yes	Completed	Team collaboration on final package: compiled all components into zip file, final document review, and submission to Canvas as per Part B instructions.

Legend

Task ID:	Each task is to have a unique ID (the ID is repeated weekly until the task is reviewed, accepted and marked as completed)
WIP Uploaded Status:	Yes, No
Task Status:	Completed, In Progress, In Review, Reassigned to (member IDs)

Note on Equal Contribution: While specific members led certain tasks, all deliverables involved genuine collaboration with input, validation, and support from the entire team, resulting in e